

QUESTIONS 4 / 8 completed

1-10 T

1-10 T

Write a Java program to create a simple thread that prints the numbers from 1 to 10.

PROGRAM EDITOR

Java

Run

Save

```
1 class NumberThread extends Thread{
2     @Override
3     public void run(){
4         for(int i=1;i<=10;i++){
5             System.out.println(i);
6             try{
7                 Thread.sleep(100);
8             }catch(InterruptedException e){
9                 System.out.println("Thread interrupted: "+e.getMessage());
10            }
11        }
12    }
13 }
14 public class Main{
15     public static void main(String[] args){
16         NumberThread thread= new NumberThread();
17         thread.start();
18     }
19 }
```

OUTPUT

INPUT

Your INPUT goes here!

QUESTIONS 4 / 8 completed

Counter-T ▼

COUNTER-T

Write a Java program to create a thread-safe counter that can be accessed by multiple threads simultaneously.

PROGRAM EDITOR

Java ▼

Run

Save

```

1 class counter{
2     private int count=0;
3     public synchronized void increment(){
4         count++;
5     }
6     public synchronized int getCount(){
7         return count;
8     }
9 }
10 public class Main{
11     public static void main(String[] args) throws InterruptedException{
12         Counter counter = new Counter();
13         Thread t1 = new Thread()->{
14             for(int i=0;i<1000;i++){
15                 counter.increment();
16             }
17         };
18         Thred t2 = new Thread()->{
19             for(int i=0;i<1000;i++){
20                 counter.increment();
21             }

```

OUTPUT

```

Compilation Error:
C:\Windows\TEMP\sandbox_192511318RC_6a76b48fd53726.83813090\counter.java:10:
error: class Main is public, should be declared in a file named Main.java
public class Main{
    ^
C:\Windows\TEMP\sandbox_192511318RC_6a76b48fd53726.83813090\counter.java:12:
error: cannot find symbol
    Counter counter = new Counter();
    ^
symbol:   class Counter

```

INPUT

Your INPUT goes here!

QUESTIONS 4 / 8 completed

M-Thread

M-THREAD

Write a Java program to create multiple threads that perform some time-consuming task, such as calculating prime numbers or sorting an array, and compare their execution times.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Arrays;
2 import java.util.Random;
3 class TaskThread extends Thread {
4     private String threadName;
5     private int size;
6     public TaskThread(String name, int size) {
7         this.threadName = name;
8         this.size = size;
9     }
10    @Override
11    public void run() {
12        System.out.println(threadName + " started.");
13        long startTime = System.currentTimeMillis();
14        int[] array = new int[size];
15        Random random = new Random();
16        for (int i = 0; i < size; i++) {
17            array[i] = random.nextInt(100000);
18        }
19        Arrays.sort(array);
20        long endTime = System.currentTimeMillis();
21    }
```

OUTPUT

```
Error: Main method not found in class TaskThread, please define the main method
as:
    public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
```



Search



QUESTIONS 4 / 8 completed

Pro-Consume

PRO-CONSUME

Write a Java program to create a producer-consumer scenario using multiple threads. The producer thread should add items to a queue, and the consumer thread should remove items from the queue and process them.

PROGRAM EDITOR

Java

Run

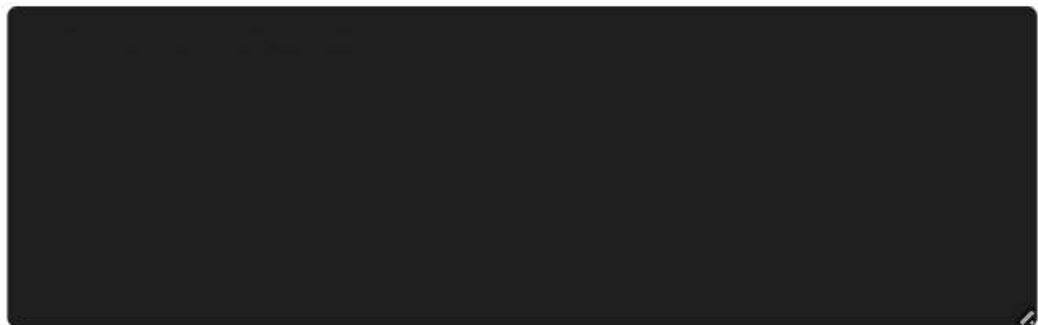
Save

```

1 import java.util.concurrent.ArrayBlockingQueue;
2 import java.util.concurrent.BlockingQueue;
3 class Producer implements Runnable{
4     private final BlockingQueue<Integer>queue;
5     public producer(BlockingQueue<Integer>queue){
6         this.queue=queue;
7     }
8     @Override
9     public void run(){
10         try{
11             for(int i=1;i<=5;i++){
12                 System.out.println("Produced items:"+i);
13                 queue.put(i);
14                 Thread.sleep(500);
15             }
16         }catch(InterruptedException e){
17             Thread.currentThread().interrupt();
18         }
19     }
20 }
21 class

```

OUTPUT



INPUT

Your INPUT
here!



Search

